

LLMs for Economics Research

A very opinionated tour

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Applied Reading Group
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Why am I giving a talk about AI?

- I am not an AI expert.
- In fact, I am not an expert in anything!
- I am, however, an *infovore*.
- I spend too much time being connected.
- So perhaps I can give you a map of things I have found useful.

This is a very opinionated map.

A great course to get started

10-202: Introduction to Modern AI

by Zico Kolter

► modernaicourse.org

Course Contents

- AI history and foundations: linear algebra and PyTorch
- Supervised learning: linear models, loss functions, gradient descent and neural networks
- Language models: attention, Transformers, tokenisation, inference and training
- Post-training: instruction tuning, reinforcement learning, RLHF and reasoning
- AI safety, security and the future of AGI

1950s–1960s: Rosenblatt's perceptron



“the machine would be the first device to think as the human brain. As do human beings, Perceptron will make mistakes at first, but will grow wiser as it gains experience”

The fracturing of AI: 80s–00s

- 1980s — neural networks trained via backpropagation, convolutional and recurrent networks
- 1997 — Deep Blue beat Garry Kasparov
- 2000s — Ads / recommendations
- 2005/2007 — DARPA Grand Challenges (autonomous driving)

The rebirth of AI: 10s–present

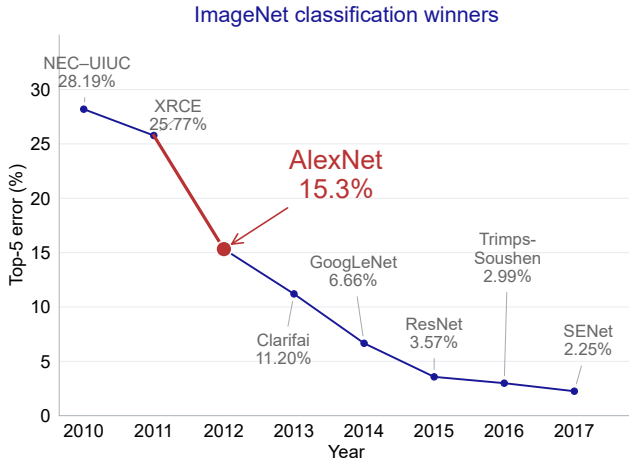
- 2012 — AlexNet (emergence of deep learning)
- 2014 — Seq2Seq (machine translation)
- 2016 — AlphaGo
- 2017 — Transformer paper
- 2022 — ChatGPT
- Present — AI everywhere

2012: AlexNet



Fei-Fei Li

► TED talk



2017: The Transformer

Attention Is All You Need

Ashish Vaswani* **Noam Shazeer*** **Niki Parmar*** **Jakob Uszkoreit*** **Llion Jones***
Google Brain Google Brain Google Research Google Research Google Research

Aidan N. Gomez* [†]
University of Toronto

Łukasz Kaiser*
Google Brain

Illia Polosukhin* [‡]

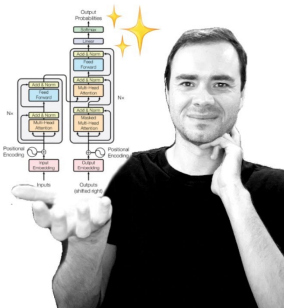
► [Original paper](#)

► [The Annotated Transformer \(PyTorch\)](#)

The architecture behind the **T** in GPT.

Understanding GPT with Andrej Karpathy

LET'S BUILD GPT.
FROM SCRATCH.
IN CODE.
SPELLED OUT.



A very clear, step-by-step coding course (about 2 hours).

► Lecture: Let's build GPT



Deep learning for solving economic models

Journal of Economic Literature 2026, 64(3), 829–875
<https://doi.org/10.1257/jel.20261794>

Deep Learning for Solving Economic Models[†]

JESÚS FERNÁNDEZ-VILLAYERDE*

The ongoing revolution in deep learning is reshaping research across many fields, including economics. Its effects are especially clear in solving dynamic economic models. These models often lack closed-form solutions, so economists have long relied on numerical methods such as value function iteration, perturbation, and projection techniques. Unfortunately, these approaches suffer from the curse of dimensionality, which makes global solutions computationally infeasible as the number of state variables increases. Deep learning offers a different approach: flexible tools that solve dynamic economic models by minimizing residuals in equilibrium conditions and that can handle high-dimensional problems. This development promises to broaden the scope of quantitative economics. I illustrate the approach using the neoclassical growth model. (JEL C45, C61, D83, O41)

The problem

- Let us suppose we want to approximate (“learn”) an unknown function:

$$y = f(\mathbf{x})$$

where y is a scalar and $\mathbf{x} = \{x_0 = 1, x_1, x_2, \dots, x_N\}$ a vector (why a constant?).

- We care about the case when N is large (possibly in the thousands!).
- Easy to extend to the case where y is a vector (e.g., a probability distribution), but notation becomes cumbersome.
- In economics, $f(\mathbf{x})$ can be a value function, a policy function, a pricing kernel, a conditional expectation, a classifier, ...

A neural network

- A neural network is an approximation to $f(\mathbf{x})$ of the form:

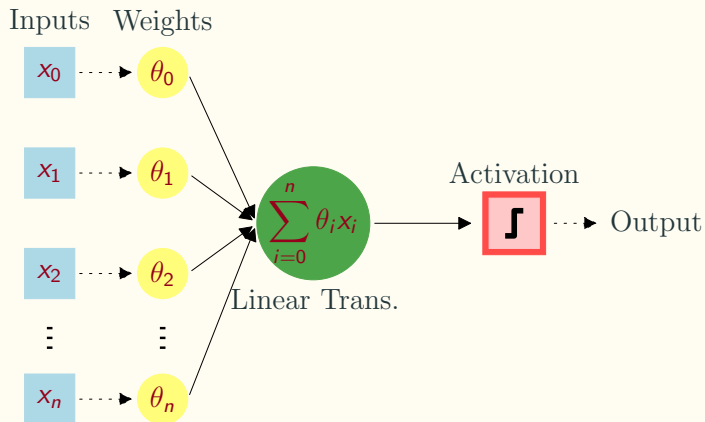
$$y \cong g^{NN}(\mathbf{x}; \theta) = \theta_0 + \sum_{m=1}^M \theta_m \phi(z_m)$$

where $\phi(\cdot)$ is an activation function and:

$$z_m = \sum_{n=0}^N \theta_{n,m} x_n$$

- The x_n 's are known as the features of the data, which belong to a feature space $\mathcal{X}$.
- The $\phi(z_m)$'s are known as the representation of the data.
- M is known as the width of the model (wide vs. thin networks).
- “Training” the network: selecting θ such that $g^{NN}(\mathbf{x}; \theta)$ is as close to $f(\mathbf{x})$ as possible given some relevant metric (e.g., the ℓ_2 norm).

Flow representation



Deep learning I

- A deep learning network is a *multilayer* composition of $J > 1$ neural networks:

$$z_m^0 = \theta_{0,m}^0 + \sum_{n=1}^N \theta_{n,m}^0 x_n$$

and

$$z_m^1 = \theta_{0,m}^1 + \sum_{m=1}^{M^{(1)}} \theta_m^1 \phi^1(z_m^0)$$

...

$$y \cong g^{DL}(\mathbf{x}; \theta) = \theta_0^J + \sum_{m=1}^{M^{(J)}} \theta_m^J \phi^J(z_m^{J-1})$$

where the $M^{(1)}, M^{(2)}, \dots$ and $\phi^1(\cdot), \phi^2(\cdot), \dots$ are possibly different across each layer of the network.

From a Few Parameters to Many Billions

- Griliches's Law: "More than five variables in a cross section yields garbage."
- Von Neumann: "With four parameters I can fit an elephant, and with five I can make him wiggle his trunk."
- GPT-3 (2020): 175 billion parameters; Llama 3.1 (2024): 405 billion.
- Qwen3.5 (2026): 397 billion parameters; 17 billion active per token.
- Kimi K2.5 (2026): 1,000 billion parameters; 32 billion active per token.

Overparametrization: But is this not a problem?

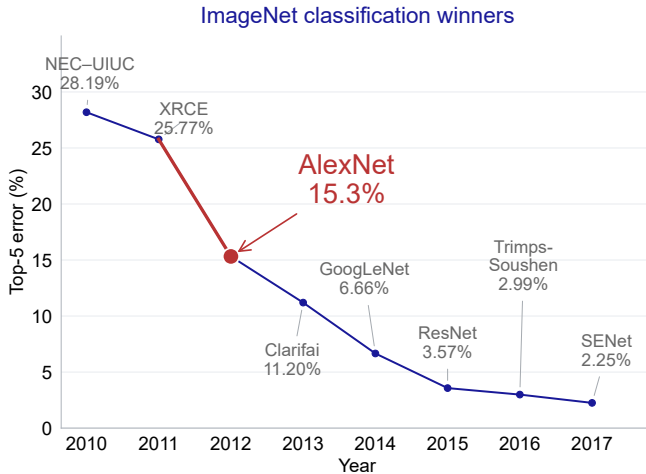
The Blessings of Overparameterization: Applications in Solving Economic Models*

Mahdi Ebrahimi Kahou[†]

Jesús Fernández-Villaverde[‡]

- As with *costates*, the aim is to return to the original economic variables.
- We can interpret and check the resulting policy or value function in that domain.
- The economic solution can be interpretable even when individual network weights are not.

It just works!!!



An LLM predicts the next token

Context $\longrightarrow$ next token $\longrightarrow$ repeat

$$P_{\theta}(t_{n+1} \mid t_1, \dots, t_n)$$

- For our purposes: a language model generates text or code, one token at a time.
- A token can be a word, part of a word, punctuation or a fragment of code.
- An embedding is the numerical representation of a token, not the token itself.
- Multimodal models also process image patches or other visual units and audio units; not necessarily individual pixels.

Add skills and tools: a powerful research assistant

LLM + skills + tools + feedback

Decide $\longrightarrow$ use a tool $\longrightarrow$ observe $\longrightarrow$ repeat

- Skills provide reusable instructions for a task or workflow.
- Tools let the system search the web, read papers and run code.
- The surrounding software executes tool calls and returns their results as context.
- The model still predicts tokens; the whole system can act, check results and revise its work.

Context: correct, complete and concise

$$\text{Performance} \sim \frac{\text{correctness}^2 \times \text{completeness}}{\text{size}}$$

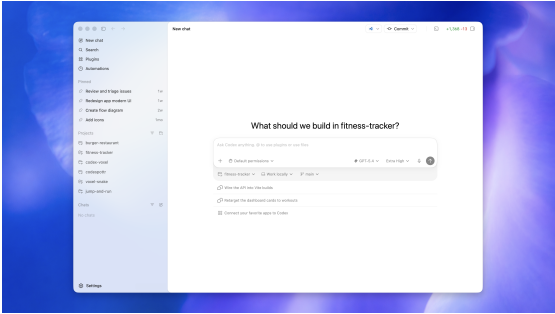
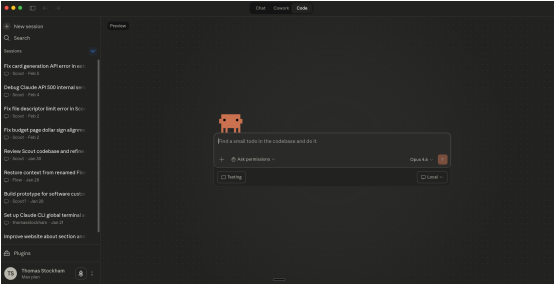
A rule of thumb for context quality, not an estimated law.

- Correctness: give reliable facts and remove obsolete or conflicting instructions.
- Completeness: supply the relevant context needed to answer the question.
- Size: each message and tool result adds tokens. More text can obscure useful information, even before the context limit.
- Keep a concise record of decisions; retrieve details when needed. At the limit, software must shorten or remove context, or reject the request.

How I use these tools

- I am not very good at prompting, I just iterate a lot!
- But I have some preferences and things that I find useful.

Claude Code and Codex



Pedro H. C. Sant'Anna

My Claude Code Setup

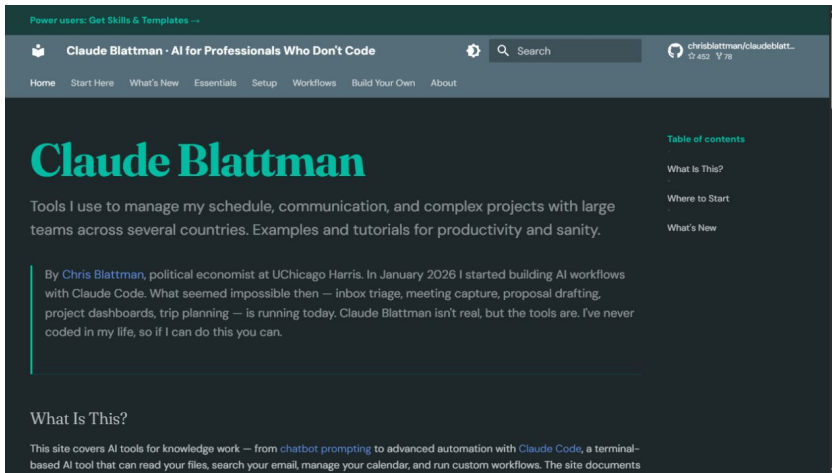
Pedro H. C. Sant'Anna · Emory University

Actively developed and maintained. This summarizes how I use Claude Code for academic work — slides, papers, data analysis, and more. Adopted by research groups across economics, energy, political science, and engineering — 15+ research groups at the March 2026 survey; the repository has since passed 2,900 forks and 1,500 stars (GitHub, 2026-08-24). A fork is not a research group — those two figures measure different things. I keep learning and updating these files.

A foundation for AI-assisted academic work using [Claude Code](#). You describe what you want — lecture slides, a research paper, a data analysis, a replication package — and Claude plans the approach, runs specialized agents, fixes issues, verifies quality, and presents results — goal-first and gate-enforced. Like a contractor who handles the entire job, with you as the auditor of the disagreements the review loop surfaces. It is deliberately not an autonomous daemon: every loop is started by you or a skill, never on its own.

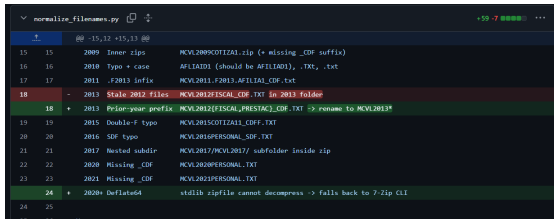
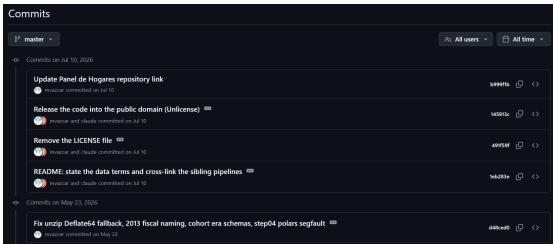
[Read the full guide](#)

Claude Blattman



Adopt modern workflows: Git and GitHub

- Register on GitHub and work with Git: track changes, review edits and collaborate.



Some skills I find really useful

- I have ADHD: short, clear responses and one step at a time.
- GOV.UK text: plain English, concise writing and clear structure.
- Scott Cunningham's Mixtape tools: Referee 2.
- No-mistakes: a workflow for checking code before delivery.

The Value of Good Questions

- I thought I had far more good questions than I actually do.
- How do I know? For simple tasks, my implementation costs have fallen to almost zero!
- The real value now lies in good questions and clever ideas.
- Almost anyone can implement a simple macro model or run a standard causal design. Knowing whether it answers the question is harder.

Verification is the bottleneck

American Economic Review 2020, 110(4): 1104–1144
<https://doi.org/10.1257/aer.20180338>

Are Ideas Getting Harder to Find?[†]

By NICHOLAS BLOOM, CHARLES I. JONES, JOHN VAN REENEN,
AND MICHAEL WEBB*

This paper applies the growth accounting of Solow (1957) to the production function for new ideas. The basic insight can be explained with a simple equation, highlighting a stylized view of economic growth that emerges from idea-based growth models:

$$\begin{array}{ccccc} \text{Economic growth} & = & \text{Research productivity} & \times & \text{Number of researchers.} \\ \text{e.g., 2\% or 5\%} & & \downarrow (\text{falling}) & & \uparrow (\text{rising}) \end{array}$$

My takeaway for AI-assisted research: more output is not the same as more verified knowledge.

Verification is the bottleneck

'coarse!

r refine

Free software; model usage costs apply.

Paid full reviews; free trial available.

- Implementation has become much more productive. Checking whether the result is right remains the bottleneck.
- A growth-theory intuition: when tasks are complements, the least improved task can constrain overall progress. My argument: verification is that bottleneck.

Weak links: verification is the bottleneck

Past Automation and Future A.I.: How Weak Links Tame the Growth Explosion

Charles I. Jones Christopher Tonetti
Stanford GSB and NBER Stanford GSB and NBER

- Faster implementation does not remove the need to check assumptions, code and results.
- My application of the weak-links idea: verification is the bottleneck in turning AI output into credible research.

Verification: data, code and journal standards

The logo for the Data and Code Availability Standard (DCAS) features the letters 'DCAS' in a bold, red, sans-serif font. The 'D' is stylized with a thick vertical stroke and a curved top. The 'C' is a simple, thick outline. The 'A' is a bold, sans-serif letter. The 'S' is a bold, sans-serif letter with a thick vertical stroke.

Data and Code Availability Standard

► Journals endorsing the standard

1. American Economic Journal: Applied Economics  
2. American Economic Journal: Economic Policy  
3. American Economic Journal: Macroeconomics  
4. American Economic Journal: Microeconomics  
5. American Economic Review  
6. American Economic Review: Insights  
7. Canadian Journal of Economics 
8. Econometrica 
9. Econometrics Journal
10. Economic Inquiry 
11. Economic Journal 
12. Journal of Economic Literature  
13. Journal of Economic Perspectives  
14. Journal of Political Economy 
15. Journal of the European Economic Association 
16. Quantitative Economics 
17. Review of Economic Studies 
18. Review of Financial Studies 
19. Theoretical Economics 

Don't be a meat proxy

Niklas Gruhn



Don't be a meat proxy

Aug 03, 2026

Too often I ask a question in Slack or leave feedback under a merge/pull request or argue with friends in a WhatsApp group and get back:

| *Claude said: [giant response verbatim]*

Please don't do this. I mean, I've done this. But I've been on the receiving end too many times now. This is not adding value. I can talk to Claude myself. It's going to be faster and I get to control the context. I don't need a meat proxy in between.